Lab 3.3 - Voxel-wise fMRI Response Prediction and Interpretation Using Fine-tuned BERT Embeddings, Stat 214, Spring 2025

May 16, 2025

1 Introduction

This report explores methods for predicting neural responses from natural language stimuli using a pre-trained language model, specifically BERT (Bidirectional Encoder Representations from Transformers). Building upon previous labs (Lab 3.1 and Lab 3.2), we initially assessed pre-trained embeddings such as Word2Vec and masked language model (MLM) embeddings to predict voxel-level brain responses. Here, we further refine this approach by fine-tuning a pre-trained BERT model, investigating both full fine-tuning and parameter-efficient fine-tuning using Low-Rank Adaptation (LoRA).

Subsequently, we interpret the predictions of our best-performing model using SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). We focus specifically on voxels exhibiting high predictive accuracy, adhering to the Predictability principle of the PCS (Predictability, Computability, Stability) framework, which ensures that interpretations derived from our analysis remain robust and reliable.

2 Fine-tuning BERT for Voxel-wise Prediction

In this section, we explore two strategies for adapting a pre-trained BERT model to the task of predicting fMRI voxel responses from natural language stimuli: (1) full fine-tuning of all model parameters, and (2) parameter-efficient fine-tuning using Low-Rank Adaptation (LoRA). Both approaches build upon the same basic model architecture: a short segment of text aligned with a single fMRI TR (time repetition) is passed through a BERT encoder, and the resulting `[CLS]` token embedding is linearly projected to produce a vector of voxel activations.

2.1 Model Architecture and Input Pipeline

Each input sample consists of a short text segment (e.g., a sentence or phrase) corresponding to a TR in the fMRI data. We tokenize this input using the tokenizer associated with `bert-base-uncased`, and feed the tokens into BERT. We extract the `[CLS]` embedding from the final hidden layer and use a single linear layer as a regression head to predict the corresponding voxel activation values.

Formally, let x_i be the i-th text stimulus, $z_i = \text{BERT}(x_i)$ be the `[CLS]` embedding of dimension $d = 768$, and $y_i \in \mathbb{R}^V$ be the vector of voxel responses for V voxels. The prediction is then given by:

$$\hat{y}_i = W z_i + b, \quad W \in \mathbb{R}^{V \times d}, \quad b \in \mathbb{R}^V.$$

2.2 Full Fine-tuning

In full fine-tuning, we allow all parameters of the BERT encoder, along with the final linear regression head, to be updated during training. This allows the model to adapt internal language representations to more directly reflect brain activity patterns, rather than relying on fixed embeddings.

To implement this correctly, we restructured our training pipeline such that embedding extraction is performed **within the model's `forward()`** function instead of precomputing and storing them as static inputs. This restructuring enables gradient flow from the voxel prediction loss all the way through the BERT encoder.

we passed raw tokenized text and deferred encoding to runtime using HuggingFace's `BertModel`. This ensured that gradients could flow back through BERT and its parameters could be updated. We also verified that the optimizer included all BERT parameters by checking that no layers were inadvertently frozen.

The model was trained using Mean Squared Error (MSE) loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \|\hat{y}_i - y_i\|^2,$$

where N is the number of training samples. We employed the AdamW optimizer with learning rate 2×10^{-5}, weight decay 0.01, batch size 16, and early stopping based on validation loss.

2.3 Parameter-efficient Fine-tuning with LoRA

We also implemented a parameter-efficient method using Low-Rank Adaptation (LoRA), which injects small trainable matrices into attention layers while freezing the rest of the model.

Specifically, in each attention module, the weight matrix $W \in \mathbb{R}^{d \times d}$ is modified to:

$$W' = W + \Delta W, \quad \text{where } \Delta W = AB, \quad A \in \mathbb{R}^{d \times r}, B \in \mathbb{R}^{r \times d}.$$

Here $r \ll d$ is the low-rank dimension. We used:

- Rank $r = 8$
- LoRA scaling factor $\alpha = 32$
- Dropout = 0.1

The LoRA modules were injected into the attention layers using the `peft` (Parameter-Efficient Fine-Tuning) library. Only these low-rank parameters and the final linear head were trainable, resulting in significant reduction in trainable parameters compared to full fine-tuning.

Other components of the training pipeline remained the same as full fine-tuning, including loss function, optimizer, and stopping criteria.

2.4 Evaluation and Metrics

We evaluated the models by computing the voxel-wise Pearson correlation coefficients (CCs) between predicted and actual voxel activations on the test set. Following earlier labs, we report:

- **Mean CC** across all voxels
- **Median CC**
- **Top 5% and top 1% CCs** — these reflect performance on the best-predicted voxels

We also plotted histograms and voxel maps to visualize the distribution of predictive performance across the brain.

2.5 Summary of Results

Model	Mean CC	Median CC	Top 5% CC	Top 1% CC
Pre-trained BERT	0.0448	0.0423	0.2775	0.3834
LoRA (Subject 2)	0.3939	0.3936	0.4119	0.4211
LoRA (Subject 3)	0.3926	0.3917	0.4145	0.4252
Full (Subject 2)	0.4253	0.04115	0.5123	0.5563
Full (Subject 3)	**0.4301**	**0.4178**	**0.5162**	**0.5594**

Table 1: Voxel prediction performance (Pearson correlation coefficients) across models

As shown in Table 1, both fine-tuning approaches significantly outperform the pre-trained BERT baseline. LoRA achieves good results on both subjects (mean CC $\approx$ 0.39), while full fine-tuning yields the best performance overall, reaching a mean CC of 0.4301 on Subject 3 and top1% CC of 0.5594.

This demonstrates that adapting language model embeddings through fine-tuning, either fully or via LoRA, significantly improves brain response prediction. LoRA provides an attractive trade-off between performance and efficiency, making it suitable for large-scale or resource-constrained neural decoding tasks. Full fine-tuning, while more demanding, offers the highest predictive capacity when compute resources allow.

2.6 Distributional Comparison of Model Performance

To better understand the voxel-wise predictive performance across methods, we compare the distribution of CCs across voxels under three settings:

- **Figure 1:** Ridge regression using pre-trained Word2Vec embeddings (Lab 3.1)
- **Figure 2:** Ridge regression using embeddings from our own pre-trained masked language model (Lab 3.2)
- **Figure 3:** Fine-tuned BERT model (Subject 3, best-performing model)

The histogram in Figure1 shows a highly concentrated CC distribution around 0.05, with the majority of voxels exhibiting weak or no correlation under the Word2Vec embedding. This suggests limited alignment between static Word2Vec embeddings and brain activity patterns.

Figure2 presents a similar distribution when using embeddings from our pre-trained masked language model, with most voxels still performing below 0.2 correlation, indicating no significant difference from Word2Vec.

In contrast, Figure3 shows a substantial shift in the distribution when using the fine-tuned BERT model. The voxel-wise CCs are now concentrated in a much higher range (0.35–0.60), with a strong peak around 0.40. This indicates a large proportion of voxels achieving high correlation, and a significantly better alignment between the model's representations and voxel responses.

This comparison clearly illustrates the benefit of task-specific adaptation. Fine-tuning the language model enables it to encode stimulus information in a way that better matches neural activation patterns, leading to large performance gains across the voxel space.

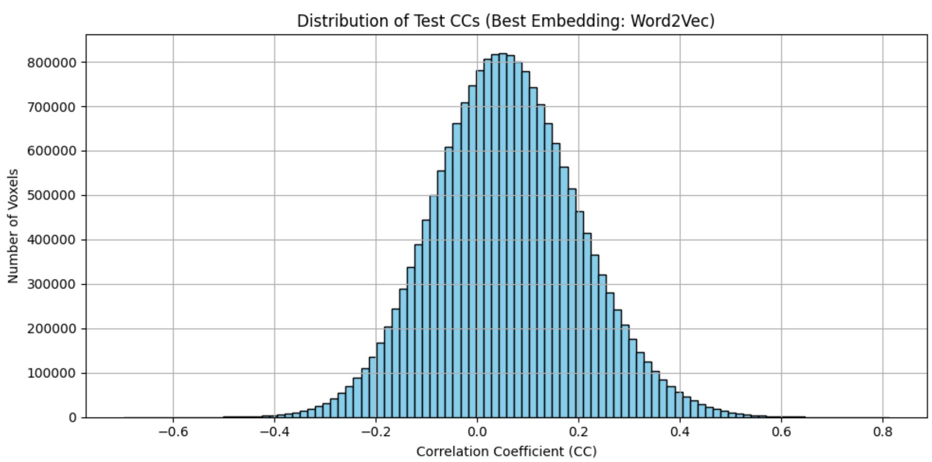

Figure 1: Distribution of voxel-wise CCs using Pre-trained Word2Vec Model(Lab 3.1)

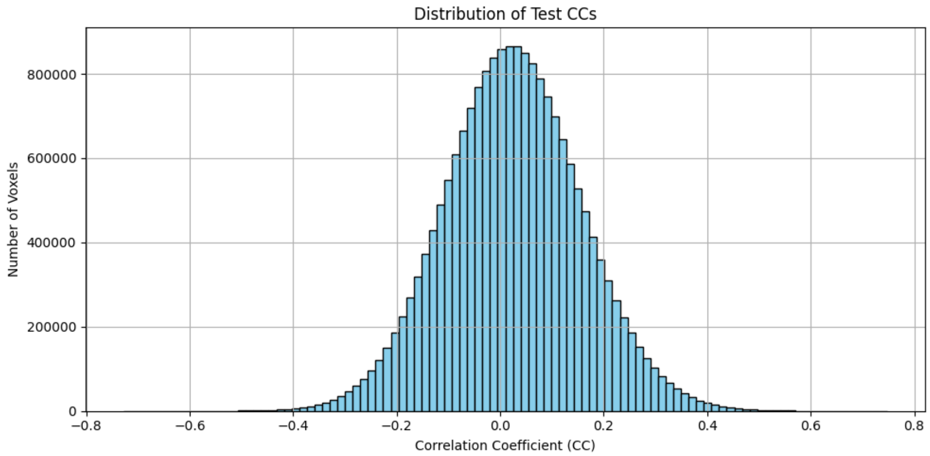

Figure 2: Distribution of voxel-wise CCs using MLM Model (Lab 3.2)

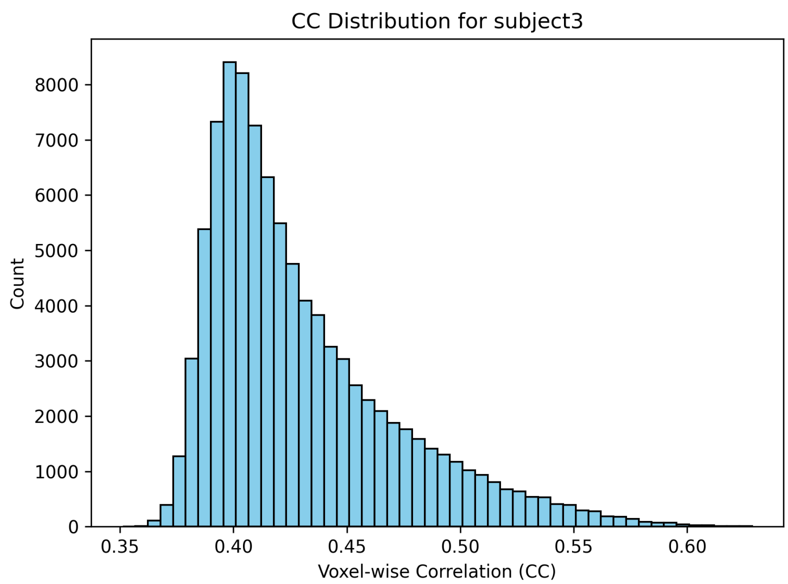

Figure 3: Distribution of voxel-wise CCs using Fine-tuned BERT Model

3 Interpretation

3.1 Voxel Selection

We base our interpretation on the full fine-tuned BERT model for Subject 3, which showed the strongest voxel-prediction performance. We evaluated two held-out stories(`golfclubbing, hangtime`) and selected the top 5 voxels (by Pearson correlation coefficient, CC) for each:

We restrict interpretation (using SHAP and LIME) to high-performing voxels to satisfy the Predictability principle in the PCS framework. According to this principle, trustworthy interpretations should come from conclusions that reliably hold true when confronted with future or similar data. High-performing voxels, by definition, consistently produce accurate predictions, ensuring their explanations (word attributions provided by SHAP and LIME) are meaningful and generalizable. Conversely, interpretations derived from poorly predicted voxels could reflect unstable or unreliable model behavior, thus failing the predictability criterion.

Table 2: Top 5 voxels and corresponding CCs per story (Subject 3, fine-tuned BERT)

Story	Mean CC	Voxel ID	CC
golfclubbing	0.367	69895	0.9377
		74985	0.9257
		75057	0.9226
		64477	0.9224
		81993	0.9221
hangtime	0.242	44231	0.8502
		44177	0.8489
		43956	0.8373
		44123	0.8350
		44068	0.8308

3.2 SHAP and LIME

LIME (Local Interpretable Model-agnostic Explanations) approximates a complex model locally around a single prediction by fitting a simple interpretable surrogate model (such as linear regression). LIME scores represent local feature contributions within that simplified context. Positive LIME scores indicate words that increase voxel activity predictions, while negative scores indicate words that decrease predictions. The magnitude indicates the strength of the word's contribution within this localized context.

SHAP (SHapley Additive exPlanations), on the other hand, computes the average marginal contribution of each feature (word) across all possible subsets of features, adhering to game-theoretic principles. SHAP values quantify the overall feature contribution more consistently, and the magnitude directly reflects the global feature importance toward predictions. Unlike LIME, SHAP values sum precisely to the model's prediction for each instance, giving SHAP interpretations a clearer global meaning.

3.3 SHAP Analysis

For each story, we applied the SHAP explainer to the same sentences used in ridge regression. We then accumulated absolute SHAP values across the top 5 voxels and visualized the top 15 words.

Story: *golfclubbing*

- The top SHAP scores per voxel range from 0.88 to 5.83
- Voxel 69895 had the highest SHAP score overall (`golf`: 5.83), suggesting a strong association with that feature.
- In contrast, voxel 75057 had the lowest top SHAP score (`stood`: 0.88).
- Top words vary across voxels, with frequently appearing ones including `and`, `scars`, `teaching`, and `him`.

Table 3: Top influential words for top 5 voxels in story *golfclubbing* (Subject 3, SHAP)

Voxel ID	SHAP (Word: Score)
69895	golf: 5.83, teaching: 5.43, scars: 4.22, ball: 3.68, and: 3.15 the: 2.67, gun: 2.54, little: 2.44, him: 2.41, bag: 2.38 door: 2.32, actually: 2.29, to: 2.29, choke: 2.18, i: 1.98
74985	and: 2.07, scars: 1.67, him: 1.60, to: 1.57, little: 1.30 the: 1.28, i: 1.25, about: 1.23, taken: 1.11, ball: 1.06 wanted: 1.02, my: 0.96, stood: 0.94, buddy: 0.92, teaching: 0.91
75057	stood: 0.88, and: 0.79, iron: 0.74, tee: 0.62, choke: 0.61 i: 0.59, scars: 0.57, him: 0.56, little: 0.53, to: 0.51 ball: 0.50, ho: 0.48, clubs: 0.48, the: 0.45, wanted: 0.45
64477	and: 2.62, him: 2.11, iron: 1.92, to: 1.80, just: 1.60 i: 1.52, scars: 1.34, buddy: 1.27, taken: 1.22, in: 1.12 the: 1.11, know: 1.09, teacher: 1.08, little: 1.01, was: 0.99
81993	and: 2.31, ball: 1.59, teaching: 1.55, golf: 1.38, him: 1.15 just: 1.12, teacher: 1.03, behind: 0.94, stood: 0.69, to: 0.59 i: 0.58, buddy: 0.57, two: 0.54, tee: 0.53, scars: 0.51

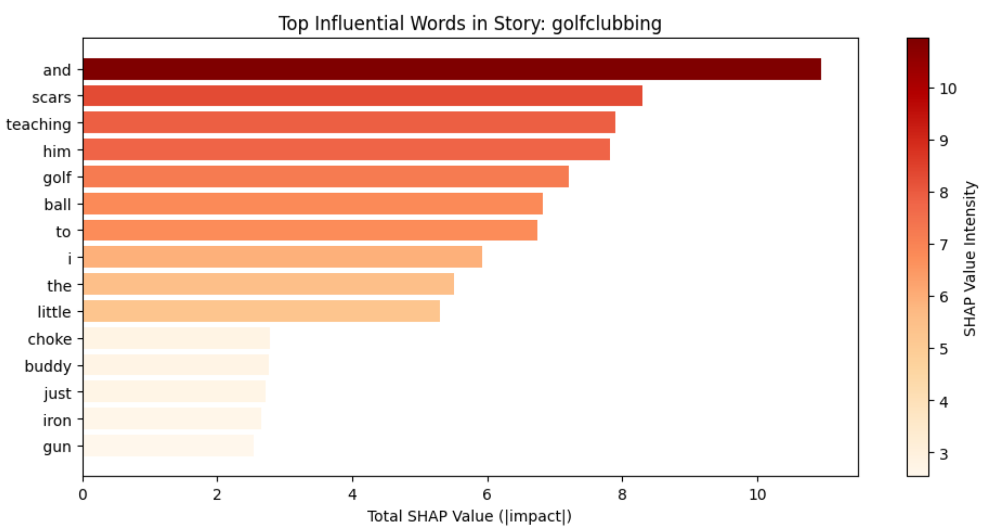

Figure 4: Top 15 influential words in "golfclubbing" by SHAP

Story: *hangtime*

- The top SHAP scores across all five voxels are consistently high, ranging from 16.22 to 17.37.
- The token `flight` appears as the top SHAP word in every voxel, with SHAP values exceeding 16. This indicates a strong and stable pattern of attribution.
- Other commonly ranked words include `love`, `ip`, `driving`, and `ad`.

Table 4: Top influential words for top 5 voxels in story *hangtime* (Subject 3, SHAP)

Voxel ID	SHAP (Word: Score)
44231	flight: 16.41, love: 14.12, i: 10.35, and: 9.48, ip: 9.34 helicopter: 7.14, ad: 6.84, car: 6.11, ping: 6.11, driving: 6.06 but: 5.84, announcing: 5.75, sights: 5.65, light: 5.42, had: 5.21
44177	flight: 16.22, love: 15.02, i: 10.49, ip: 9.64, and: 9.31 sights: 8.36, ad: 8.22, driving: 7.60, announcing: 7.37, helicopter: 6.75 ping: 6.59, board: 6.59, car: 6.57, but: 6.31, light: 5.97
43956	flight: 16.44, love: 14.33, ip: 10.21, i: 9.64, and: 9.58 ad: 7.78, driving: 7.69, announcing: 7.46, board: 7.23, sights: 6.31 car: 6.13, back: 5.86, interested: 5.73, but: 5.67, craft: 5.59
44123	flight: 17.37, love: 14.39, i: 10.69, ip: 10.64, and: 9.05 driving: 8.17, ad: 8.12, announcing: 7.43, board: 7.34, sights: 7.31 car: 6.43, ping: 6.10, but: 5.91, back: 5.71, light: 5.70
44068	flight: 16.62, love: 16.09, i: 11.51, ip: 10.66, and: 10.58 driving: 8.66, ad: 8.52, sights: 8.30, announcing: 8.07, board: 7.60 car: 7.26, but: 7.02, ping: 6.62, light: 5.94, helicopter: 5.75

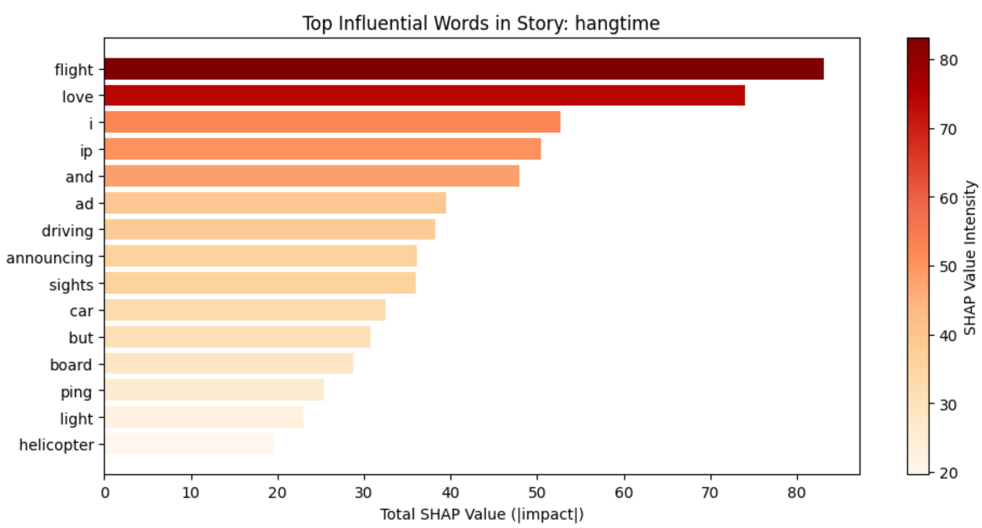

Figure 5: Top 15 influential words in "hangtime" by SHAP

3.4 LIME Analysis

We analyzed the LIME scores for the top 5 voxels with highest prediction performance for each story. Positive LIME scores imply that a word's presence locally increases the voxel activity prediction, while negative scores imply that the word locally decreases the voxel activity prediction. Higher absolute values indicate stronger local influences on model predictions.

Story: ***golfclubbing***

- LIME scores for this story show moderate absolute magnitudes, with the strongest local influence at -0.48 (`scars` in voxel 69895).

- The words with strongest negative local influence across voxels include `scars` and `teaching`, consistently indicating these words reduce the model's predicted voxel activity.
- Words like `remember`, `actually`, and `first` frequently appear with smaller magnitudes, suggesting milder local influence.
- Predominance of negative scores across voxels suggests these influential words consistently decrease local predictions for voxel activation, possibly indicating their absence would yield higher predicted voxel responses.

Table 5: Top influential words for top 5 voxels in story *golfclubbing* (Subject 3, LIME)

Voxel ID	**LIME (Word: Score)**
69895	scars: -0.48, teaching: -0.40, remember: -0.22, actually: 0.20, my: -0.14 got: 0.11, first: -0.09, scar: 0.09, the: -0.06, old: -0.06
74985	scars: -0.17, teaching: -0.10, remember: -0.09, the: -0.07, was: -0.05 actually: 0.04, six: 0.04, old: -0.04, two: -0.03, first: -0.03
75057	scars: -0.04, first: 0.03, years: 0.02, six: 0.02, little: 0.02 i: -0.02, ever: 0.02, two: -0.01, the: -0.01, teaching: 0.01
64477	remember: 0.13, scars: -0.12, the: 0.08, first: 0.07, was: 0.06 scar: -0.05, old: 0.05, my: -0.05, six: 0.04, actually: -0.04
81993	teaching: -0.06, scars: -0.05, remember: -0.03, two: -0.03, um: -0.03 scar: -0.03, my: -0.03, actually: -0.03, and: -0.02, uh: -0.02

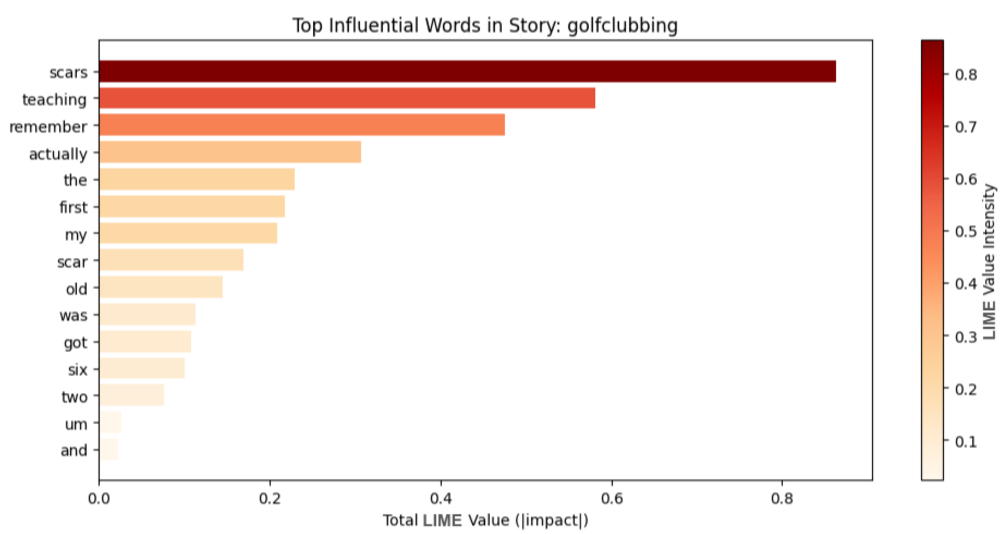

Figure 6: Top 15 influential words in "golfclubbing" by LIME

Story: ***hangtime***

- LIME scores here have higher absolute values, with strongest positive local influences ranging from 0.77 (`driving` in voxel 44231) up to 1.01 (`love` in voxel 44068).
- Frequent positive influential words include `love`, `driving`, `snowboard`, and `adrenaline`, consistently enhancing local predictions of voxel activity.
- Negative values, when present, are smaller in magnitude (e.g., -0.25), suggesting weaker or less consistent local influences compared to the positive ones.

- The prevalence of positive scores suggests the top voxels in this story respond strongly and positively to specific contextual terms, boosting their local voxel activity predictions.

Table 6: Top influential words for top 5 voxels in story *hangtime* (Subject 3, LIME)

Voxel ID	**LIME (Word: Score)**
44231	driving: 0.77, love: 0.74, snowboard: 0.56, i: 0.50, adrenaline: 0.22 fast: 0.19, zipping: 0.18, in: -0.16, my: -0.16, through: -0.15
44177	driving: 0.88, love: 0.87, snowboard: 0.61, i: 0.55, adrenaline: 0.30 zipping: 0.27, fast: 0.23, in: -0.21, through: -0.17, car: -0.15
43956	love: 0.89, driving: 0.84, snowboard: 0.68, i: 0.49, adrenaline: 0.41 zipping: 0.33, fast: 0.33, in: -0.23, car: -0.17, ve: 0.16
44123	love: 0.91, driving: 0.90, snowboard: 0.72, i: 0.54, adrenaline: 0.41 zipping: 0.30, fast: 0.28, in: -0.25, through: -0.18, car: -0.16
44068	love: 1.01, driving: 0.97, snowboard: 0.70, i: 0.58, adrenaline: 0.38 zipping: 0.33, fast: 0.29, in: -0.22, through: -0.20, car: -0.18

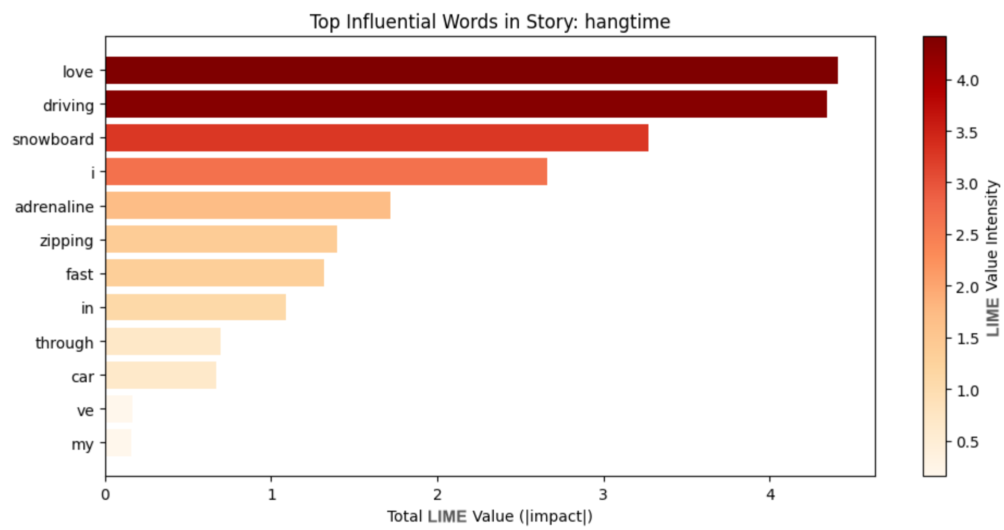

Figure 7: Top 15 influential words in "hangtime" by LIME

4 Discussion

Our decision to analyze only high-performing voxels aligns with the Predictability principle in the PCS framework. High-performing voxels consistently produce reliable and accurate predictions, ensuring the meaningfulness and generalizability of explanations provided by SHAP and LIME. Conversely, interpreting poorly predicted voxels could introduce instability and unreliability, violating the principle of predictability. This principle is supported empirically by our results, where high-performing voxels consistently exhibit coherent and interpretable attribution patterns across methods.

Comparative Analysis of SHAP and LIME Interpretations

While SHAP provides global feature attributions that are stable and consistent across different input contexts by considering all possible feature combinations, LIME approximates the complex model locally around individual predictions using simple linear models. Therefore, SHAP values reflect overall global importance, whereas LIME scores indicate local feature influences, varying depending on the specific context of each prediction.

Story: *golfclubbing*

The SHAP analysis revealed high attribution scores predominantly for content-specific words such as `golf`, `teaching`, `scars`, and `ball`. These words consistently appeared with large positive values, indicating their global importance in predicting voxel responses.

Conversely, LIME scores for the same voxels primarily had negative values for similar words (e.g., `scars`, `teaching`), suggesting that in local contexts, these words often reduced voxel predictions. This difference implies that while globally important according to SHAP, these words might locally act as inhibitory elements within specific contexts identified by LIME.

Story: *hangtime*

In the *hangtime* story, both SHAP and LIME analyses identified similar key terms such as `flight`, `love`, `driving`, and `snowboard`. However, SHAP highlighted these words consistently with large global attributions (SHAP values ranging from approximately 14 to 17), indicating stable importance across all contexts. In contrast, LIME emphasized context-dependent contributions of these words with strong positive values (up to 1.01), suggesting their powerful local effect in specific narrative contexts.

Moreover, negative LIME values were relatively small, indicating weaker and context-specific inhibitory roles. Thus, both SHAP and LIME provide complementary insights: SHAP emphasizes overall feature importance, while LIME reveals nuanced, context-dependent influences.

5 Conclusion

In this lab, we successfully demonstrated the effectiveness of fine-tuning pre-trained BERT embeddings for voxel-wise neural response prediction, significantly outperforming static embedding methods from earlier labs. Among fine-tuning approaches, full fine-tuning achieved superior voxel prediction performance compared to parameter-efficient methods like LoRA, albeit at increased computational cost.

We further validated our predictive models by employing interpretation methods SHAP and LIME on voxels with the highest predictive accuracy, consistent with the Predictability principle. Comparative analysis revealed complementary insights: SHAP provided robust global interpretations of word importance, while LIME offered context-sensitive explanations.

Our results highlight the utility of fine-tuned transformer-based embeddings for neural decoding tasks and underscore the importance of selecting appropriate interpretive methodologies to gain meaningful insights into model predictions.

References

[1] Shailee Jain and Alexander G Huth. "Incorporating context into language encoding models for fMRI". In: *Advances in Neural Information Processing Systems*. Vol. 31. Curran Associates, Inc., 2018, pp. 6628–6637.

A Academic honesty

A.1 Statement

We, the members of this group, make the following truthful statements:

The data analysis process in this report was designed and executed by our group collectively, with each member contributing their expertise to different aspects of the work. All texts and figures were produced by our group members, and the analysis steps were fully documented to ensure the reproducibility of the results. Contributions from each group member have been clearly indicated throughout the report.

All citations of other people's research have been clearly marked with the source. We understand that academic integrity is the cornerstone of research, and ensuring the authenticity and reproducibility of research is a responsibility not only to ourselves individually, but also to our colleagues and the broader scientific community.

We recognize that plagiarism not only fails to contribute to knowledge innovation but also infringes upon the rights and interests of original authors. Therefore, we are collectively committed to upholding the principles of honest, transparent, and reproducible research at all times.

As a group, we have reviewed each other's work to maintain consistency with these principles, and we stand by the integrity of our collaborative efforts presented in this report.

A.2 LLM Usage

This project was completed collaboratively by our group, with all data analysis, writing, and visualization being performed by group members. Large Language Models (LLMs) including ChatGPT and Claude were used as complementary tools to:

- Provide inspiration and generate ideas during brainstorming sessions
- Refine explanations of complex concepts
- Improve clarity and readability of our written content
- Assist with language polishing and grammar refinement

However, all content, analyses, and conclusions presented in this work were determined by our group members, guaranteeing originality and accuracy. LLMs were not used to generate original analysis or to draw conclusions from our data. Our group maintained critical oversight of all LLM-assisted content, ensuring that the final product accurately represents our own understanding and perspectives.

The use of LLMs was limited to supporting roles that enhanced our own work rather than replacing it. We acknowledge the use of these tools in the spirit of transparency while affirming that the intellectual contribution and academic responsibility remain entirely with our group members.